



Koneru Lakshmaiah Education Foundation

(Deemed to be University estd. u/s. 3 of the UGC Act, 1956)

Off-Campus: Bachupally-Gandimaisamma Road, Bowrampet, Hyderabad, Telangana - 500 043.

Phone No: 7815926816, www.klh.edu.in

DEPARTMENT OF CSE

PROJECT ABSTRACT SUBMISSION

COURSE: ENGINEERING CAPSTONE PROJECT – 1(23IE4053R/23IE4053A) A.Y. 2026 - 2027

TITLE OF THE PROJECT	AI-Based Smart Bug Prediction System	
	S. No.	University ID
		Name
	1	2420030012
	2	2420030014
	3	2420030368
	4	2420030573
Name of the Guide		BHAVYA

Abstract

Software defects are one of the major challenges in software development, often leading to increased development time, higher maintenance costs, and reduced software reliability. The AI-Based Smart Bug Prediction System aims to predict potential software bugs at an early stage by analyzing software project data using Artificial Intelligence techniques. The proposed system uses historical software metrics and machine learning algorithms to identify modules that are more likely to contain defects before testing and deployment. By providing early predictions, developers can focus their testing efforts on high-risk areas, reducing debugging time and improving software quality. The system also presents prediction results through a simple and user-friendly interface, enabling efficient decision-making during software development. This project demonstrates how AI can be integrated into software engineering practices to enhance productivity, reduce costs, and support the development of reliable software systems.